

Makoto Ishihara

- Ph.D. Student

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Education

- **Apr. 2025 – Present, Ph.D. Student,**

Department of Electronics and Electrical Engineering, Keio University
Supervisor: Prof. Masahiro Takeoka

- **Apr. 2023 – Mar. 2025, M.E.,**

Department of Electronics and Electrical Engineering, Keio University
Supervisor: Prof. Masahiro Takeoka
Thesis title: Long-distance quantum key distribution protocols with single-photon interference

- **Apr. 2019 – Mar. 2023, B.E.,**

Department of Electronics and Electrical Engineering, Keio University
Supervisor: Prof. Masahiro Takeoka
Thesis title: Research on capacity of quantum channels with entanglement sources

Work Experience

- **Mar. 2026 – Present, Long-term internship,**

Photonic Quantum Information (PQI), Center for Quantum Technology, Korea Institute of Science and Technology
Supervisor: Prof. Yong-Su Kim

- **Sep. 2025 – Nov. 2025, Guest Ph.D. student,**

Quantum Physics and Information Technology (QPIT), Department of Physics, Technical University of Denmark

Supervisor: Prof. Ulrik L. Andersen

Publications

- **M. Ishihara***, C. R. i Carceller*, K. N. Wilkinson*, C. A. Breum*, T. Gehring and J. B. Brask ,
Quantum randomness certification with untrusted measurements and few probe states, arXiv:2210.13608 (* equal contribution).
- M. Takeoka and **M. Ishihara**, “Quantum Key Distribution System Implementation and Devices,” in Part 4: Quantum Communication, in Handbook of Quantum Innovation, edited by Y. Arakawa (NTS, Tokyo, 2026), pp. 504-513 (in Japanese).
- **M. Ishihara**, W. Roga, U. L. Andersen and M. Takeoka, Multipartite device-independent quantum key distribution using W states, arXiv:2603.00937.
- **M. Ishihara**, A. Brendan, W. Roga, U. L. Andersen and M. Takeoka, Long-distance device-independent quantum key distribution with standard optics tools, Optica Quantum 3, 535 (2025).
- **M. Ishihara**, A. J. E. Bjerrum, W. Roga, J. B. Brask, U. L. Andersen and M. Takeoka, Long-Distance Device-Independent Conference Key Agreement, Physical Review A 112, 032429 (2025).
- Y. Ueda, **M. Ishihara**, W. Roga and M. Takeoka, Private quantum network sensing with efficient multi-partite entanglement distribution via lossy channels, arXiv:2505.10148.

Conferences

- **M. Ishihara**, W. Roga and M. Takeoka, Enhancing noise robustness in device-independent conference key agreement using the asymmetric parity-CHSH inequality, Conference on

Quantum Information (CQI) 2026, Seoul, Republic of Korea, Jul. 2026 (Poster).

- **M. Ishihara**, W. Roga and M. Takeoka, Bell inequalities tailored for W states and their applications, QCrypt2025, Hainan, China, Aug. 2025 (Poster)
- **M. Ishihara**, A. Brendan, W. Roga and M. Takeoka, Long-distance device-independent quantum key distribution with displacement-based measurement, AQIS 2025, Hong-Kong, Aug. 2025 (Poster)
- **M. Ishihara**, W. Roga and M. Takeoka, Long distance device-independent quantum cryptography with W states, Quantum Innovation 2025, Osaka, Japan, Jul. 2025 (Poster)
- **M. Ishihara**, W. Roga and M. Takeoka, Device-independent conference key agreement with mild detection efficiency, Quantum Innovation 2024, Tokyo, Japan, Oct. 2024 (Poster)
- **M. Ishihara**, A. J. E. Bjerrum, W. Roga, J. B. Brask, U. L. Andersen and M. Takeoka, Long-distance device-independent conference key agreement, QCrypt2024, Vigo, Spain, Sep. 2024 (Poster)
- **M. Ishihara**, W. Roga and M. Takeoka, Twin-field-like-continuous-variable quantum key distribution with highly non-Gaussian state, ALPS2024, Yokohama, Japan, Apr. 2024 (Talk)
- **M. Ishihara**, W. Roga and M. Takeoka, Long distance continuous-variable quantum key distribution with discrete-modulated coherent state and two-mode squeezed vacuum, Quantum Innovation 2023, Tokyo, Japan, Nov. 2023 (Poster)

Awards & Funding

- **Apr. 2025 – Mar. 2028, Support for Pioneering Research Initiated by the Next Generation, Japan Science and Technology Agency**
Research topic: Long-distance and high-performance device-independent multipartite quantum key distribution with W state
7.5 M yen for three years
- **Apr. 2025 – Mar. 2027, Research Encouragement Scholarship for Graduate Students, Keio**

University

1 M yen for two years

- **Mar. 2023, Keio Engineering Foundation Award, Keio Engineering Foundation**

Recognized for academic excellence during undergraduate studies

Teaching

- **Apr. 2025 – Sep. 2025, Teaching Assistant,**

Department of Electronics and Electrical Engineering, Keio University
Quantum Engineering

- **Apr. 2024 – Sep. 2024, Teaching Assistant,**

Department of Electronics and Electrical Engineering, Keio University
Quantum Engineering

- **Apr. 2023 – Sep. 2023, Teaching Assistant,**

Department of Electronics and Electrical Engineering, Keio University
Quantum Engineering